Verification plan & Subroutines

Requirements for all testbenches:

- Create a verification plan -> Use this verification plan as a template
- Create a self-checking testbench either by calculating the correct result as done in the class or creating a reference in the testbench that generates the expected result to check the functionality.
- Create a task for checking the normal operation of the design.
- Create a task to reset and check the reset functionality.
- Create counters to keep track of the correct count and error count. Display them by the end of your testbench.

Deliverables:

One PDF file having screenshots of the following:

1. Design
    a. If the design has bugs, then fix them and mention what you have fixed
2. Verification Plan
3. Testbench
4. Do file
5. Coverage report text file
    a. Add this option (-du <design_module_name>) to the "coverage save" command to save only the coverage of the design and exclude the testbench
1. **Clear and neat** QuestaSim waveform snippets showing the functionality of the design with **each verification plan label (column A) mentioned in your verification plan**
2. Branch, statement and toggle coverage report snippets with justification if you could not reach 100% coverage for the designs and add code coverage exclusions to reach 100% coverage (watch the demo shared on Google classroom to learn more about coverage exclusions)

Questions:

1) Add more test vectors to the adder testbench done in the class to reach 100% code coverage.

2) Verify the functionality of the following 4-to-2 priority encoder The design consists of 3 inputs which are the following:

- 4-bit input D
- clk
- synchronous active high rst which resets the outputs to 0

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The design has 2 output ports (2-bit output Y and 1-bit output valid) which update with the positive edge of clock. Output valid is set to 1 when more than one input line is high. If all the inputs are '0', then valid output is zero. In this case, the output Y is considered as don't care conditions denoted by 'X'.

D3	D2	D1	D0	Y1	Y0	valid
0	0	0	0	X	X	0
1	0	0	0	0	0	1
X	1	0	0	0	1	1
X	X	1	0	1	0	1
X	X	X	1	1	1	1

Since the design is simple, it is required to perform exhaustive verification which means generating all possible input combinations for the input D and making sure that the outputs are correct. Note that you verification plan can consists of 2 rows, the first row is checking the reset and the second row can mention that you will exhaustively verify all possible inputs combination.

3) ALU design will be provided and has the following characteristics.

- Reset which resets output C to 0.
- 4-bit signed inputs, A and B
- 5-bit registered signed output C
- 4 op-codes
    - add
    - sub (A-B)
    - bitwise invert input A
    - reduction OR input B

Assume the following encoding of the opcodes.

Opcode	**Encoding**
Add	2'b00
Sub	2'b01
bitwise invert input A	2'b10
reduction OR input B	2'b11

Perform directed verification as we did in class with the signed adder by applying test vectors that hit the boundary cases (extreme values) for both addition and subtraction. Also, consider how to verify the bitwise invert operation on port A and the reduction OR on port B to ensure the output correctly processes the intended input. Finally, at the end of the testbench, select two values for A and B and apply all opcodes to them.

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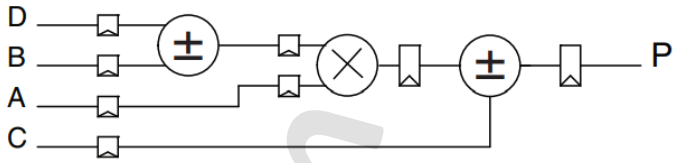
4) Verify the functionality of a simplified unsigned DSP block (DSP48A1).



Port	Type	Width
A	Input	18
B	Input	18
C	Input	48
D	Input	18
clk	Input	1
rst_n	Input	1
P	Output	48

Parameters:

1. OPERATION: take 2 values either “ADD” or “SUBTRACT”, Default value “ADD”

Notes:

- All registers are positive-edge triggered with an asynchronous active-low reset.
- The DSP will be used only in addition mode.
- Stimulus: To simplify the output checking, drive the DSP inputs every 4 clock cycles and check the P output.
- Verification Strategy:
    1. After reset verification, use a for loop to randomize the inputs.
    2. The for loop should:
        - Randomize the inputs.
        - Call the check_result task.
    3. The check_result task should:
        - Wait for 4 clock cycles and then compare the DSP output (P) with a golden model (a reference model in the testbench).
    4. Verification plan can have only 2 rows, one checking for reset and the other is checking the normal operation out of reset using randomization.
- Golden Model:
    - Implement a 3-line reference model in the testbench that mirrors the DSP operations shown in the provided figure to get the expected P and compare it with the DUT P output.

5) Watch this introductory video for OOP in SystemVerilog.

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